

Fundamentals of Statistical Learning - Unit 1

Master's Degree in Data Science, Sapienza Università di Roma
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Reference Bibliography

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Academic Year 2026/2027

Disclaimer

This document contains notes for the **Fundamentals of Statistical Learning - Unit 1** course, given by **Professor Brutti** as part of the Master's Degree in Data Science at **Sapienza Università di Roma** during the **first** semester of Academic Year **2026/2027**.

Note: These notes may not cover all details of each lecture, so it is highly suggested to consult the Professor's official resources and materials.

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Chapter 1

Defining probability

1.1 Introducing a qualitative idea of probability

Probability is the field of mathematics that studies events and how likely they are to occur in order to describe relationships or patterns in existing models.

Starting from a model's data generation process, probability aims to investigate the properties of the observed outcomes, whereas inference makes use of the observed outcomes to study the underlying data generation process.

Example: Diffusion models

Diffusion models are a type of generative model that create new data by gradually reversing a noising process.

From a mathematical perspective, a diffusion model can be described using a *Markovian Direct Acyclic Graph* whose nodes represent the observable or latent variables involved in the process and whose edges denote the transition probabilities of the model.

1.2 Formalizing probability using set theory

It is possible to formalize probabilistic experiments using set theory.

In fact, starting from a sample space Ω , which represents the set of all possible outcomes, an event A on Ω can be defined as the set of all favourable outcomes in which the event does occur.

Example: Tossing a coin

Suppose you toss a coin: it is possible to construct a random experiment by defining the sample space $\Omega = \{Heads, Tails\}$ and considering the event $A = \{Heads\}$.

1.2.1 Set operations on probability spaces

Given two events A and B on the same sample space Ω , it is possible to define three main operations:

- **Union:** The set $A \cup B = \{\omega \in \Omega : \omega \in A \text{ or } \omega \in B\}$ corresponds to the event in which at least one of A or B , or both, occurs.
In particular, it is always the case that $A \cup \Omega = \Omega$ and $A \cup \emptyset = A$.
- **Intersection:** The set $A \cap B = \{\omega \in \Omega : \omega \in A \text{ and } \omega \in B\}$ corresponds to the event in which both A and B occur.
In particular, it is always the case that $A \cap \Omega = A$ and $A \cap \emptyset = \emptyset$.
- **Complement:** The set $A^C = \Omega \setminus A = \{\omega \in \Omega : \omega \notin A\}$ corresponds to the event in which A does not occur.
In particular, it is always the case that $A \cup A^C = \Omega$ and $A \cap A^C = \emptyset$.

Additionally, given a sequence of events $\{A_1, \dots, A_n\}$ on the same sample space Ω , De Morgan's laws allow to switch between the union and intersection operations through the following identities:

$$\left(\bigcup_{i=1}^n A_i \right)^C = \bigcap_{i=1}^n A_i^C \text{ and } \left(\bigcap_{i=1}^n A_i \right)^C = \bigcup_{i=1}^n A_i^C$$

1.2.2 Sample space partitions

A sequence of events $\{A_1, \dots, A_n\}$ is said to form a partition on a sample space Ω if the following conditions are satisfied:

1. $A_1 \cup \dots \cup A_n = \Omega$, typically requiring $A_i \neq \emptyset \forall i = 1, \dots, n$ as well.
2. The sets in the sequence are pairwise disjoint, meaning that $A_i \cap A_j = \emptyset \forall i \neq j$.

Generally speaking, partitions come in handy to explain the probability of an event by decomposing it as a union of disjoint sets with respect to a sequence of mutually exclusive hypotheses.

1.3 Probability measure functions

While several definitions of probability exist, Kolmogorov's axiomatic definition is the most commonly used when defining probability measures.

According to this definition, a probability measure function $\mathbb{P} : \mathcal{P}(\Omega) \rightarrow [0, 1]$ assigns to each event $A \subseteq \Omega$ a probability $\mathbb{P}(A)$ such that:

1. $\mathbb{P}(A) \in [0, 1] \forall A \subseteq \Omega$.
2. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
3. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, with $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if and only if $A \cap B = \emptyset$.
This last axiom can be generalized for a sequence of n events through the inclusion-exclusion principle, which states that

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) \right)$$

Example: Checking polynomial identities using a randomized algorithm

Consider two polynomials of degree d :

$$F(x) = \prod_{i=1}^d (x - a_i) \text{ and } G(x) = \sum_{i=1}^d c_i x^i$$

A simple way of checking whether $F(x) \equiv G(x)$ involves expanding $F(x)$ into its canonical form and compare the resulting coefficient with those of $G(x)$, although this algorithm would require performing $\Theta(d^2)$ operations.

Knowing that, for any randomly chosen point x_0 , evaluating $F(x_0)$ and $G(x_0)$ requires $O(d)$ time, it is possible to cut down complexity using a randomized algorithm:

Algorithm 1: Randomized Algorithm for Polynomial Identities.

Data: $F(x), G(x)$

Result: Whether $F(x) \equiv G(x)$.

Generate r uniformly at random from a set $\mathcal{I} = \{1, \dots, 100d\}$;

Compute $F(r)$ and $G(r)$;

if $F(r) = G(r)$ **then**

$F(x) \equiv G(x)$;

else

$F(x) \not\equiv G(x)$;

However, the randomized nature of this algorithm can result in a wrong evaluation.

In fact, while $F(x) \equiv G(x)$ results in $F(r) = G(r) \forall r$, it might be the case that, for some r , $F(r) = G(r)$ even if $F(x) \not\equiv G(x)$: this implies that the algorithm has one-sided error as it can make mistakes only when deciding that $F(x) \equiv G(x)$.

Therefore, define the event

$$E = \{F(r) = G(r) \text{ but } F(x) \not\equiv G(x)\}$$

This event occurs if and only if r is a root of the polynomial $H(x) = F(x) - G(x)$: since $H(x)$ will always have degree at most d , it will have at most d roots, allowing to bound the error probability as follows:

$$\mathbb{P}(E) = \frac{|\{r \in \mathcal{I} : r \text{ is a root of } H(x)\}|}{|\mathcal{I}|} \leq \frac{d}{100d} = \frac{1}{100}$$

As a consequence to Kolmogorov's axiomatic definition of probability, every probability space $(\Omega, \mathbb{P})$ satisfies the property that $\mathbb{P}(A^C) = 1 - \mathbb{P}(A) \forall A \subseteq \Omega$.

Proof:

Consider the partition $\Omega = A \cup A^C$ and apply Kolmogorov's axioms.

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^C) = \mathbb{P}(A) + \mathbb{P}(A^C) \Rightarrow \mathbb{P}(A^C) = 1 - \mathbb{P}(A)$$

1.3.1 Probability bounds

Boole's inequality states that, for every sequence of events $\{A_1, \dots, A_n\}$, it holds that

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i)$$

Proof:

This inequality can be proved by induction on the length of the sequence.

- For the base case, give by $n = 1$, the inequality is trivially true as $\mathbb{P}(A_1) \leq \mathbb{P}(A_1)$.
- Assuming the inequality is true for some n , the goal is to show that it is true for $(n + 1)$ as well.

Reformulate $A_1 \cup \dots \cup A_{n+1} = (A_1 \cup \dots \cup A_n) \cup A_{n+1}$ and let

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^n A_i\right) \cup A_{n+1}\right) = \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) + \mathbb{P}(A_{n+1}) - \mathbb{P}\left(\left(\bigcup_{i=1}^n A_i\right) \cap A_{n+1}\right) \\ &\stackrel{\mathbb{P}(\cdot) \geq 0}{\leq} \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) + \mathbb{P}(A_{n+1}) \stackrel{\text{hp. ind.}}{\leq} \sum_{i=1}^n \mathbb{P}(A_i) + \mathbb{P}(A_{n+1}) = \sum_{i=1}^{n+1} \mathbb{P}(A_i) \end{aligned}$$

Having proved the base case and the inductive case, this inequality must be true $\forall n \in \mathbb{N}$, which implies that probability measure functions are always sub-additive.

As a consequence to Boole's inequality, Bonferroni's inequality states that, for every sequence of events $\{A_1, \dots, A_n\}$, it must be the case that

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) \geq \sum_{i=1}^n \mathbb{P}(A_i) - (n - 1)$$

Proof:

Reformulate the joint probability of the events using De Morgan's laws.

$$\begin{aligned} \mathbb{P}\left(\bigcap_{i=1}^n A_i\right) &= 1 - \mathbb{P}\left(\left(\bigcap_{i=1}^n A_i\right)^C\right) \stackrel{\text{De Morgan}}{=} 1 - \mathbb{P}\left(\bigcup_{i=1}^n A_i^C\right) \\ &\stackrel{\text{Boole}}{\geq} 1 - \sum_{i=1}^n \mathbb{P}(A_i^C) = 1 - \sum_{i=1}^n (1 - \mathbb{P}(A_i)) \\ &= 1 - \left(\sum_{i=1}^n 1 - \sum_{i=1}^n \mathbb{P}(A_i)\right) = \sum_{i=1}^n \mathbb{P}(A_i) - (n - 1) \end{aligned}$$

Chapter 2

Conditional probability

2.1 Understanding conditional probability

Given a sample space Ω , the conditional probability that event A occurs knowing that event B was observed can be defined as follows:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \text{ with } \mathbb{P}(B) > 0.$$

Example: Measles and red spots

Consider the two events

$$A = \{\text{I have measles}\} \text{ and } B = \{\text{I have red spots on my face}\}$$

Knowing that red spots are a common symptom of measles, it will be the case that $\mathbb{P}(B|A) = 1$. However, the same thing cannot be said about $\mathbb{P}(A|B)$ as red spots might be caused by other external factors, such as eating hot food or getting a scratch.

From a broader perspective, a conditional probability distribution can be described using a probability measure function that satisfies the following properties:

1. $\mathbb{P}(A|B) \in [0, 1] \forall A \subseteq \Omega$.
2. $\mathbb{P}(\Omega|B) = 1$.
3. For any sequence of disjoint events $\{A_1, \dots, A_n\}$, it is always the case that

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i|B\right) = \sum_{i=1}^n \mathbb{P}(A_i|B)$$

As a consequence, it is always the case that $\mathbb{P}(A^C|B) = 1 - \mathbb{P}(A|B) \forall A \subseteq \Omega$.

Proof:

Let $B = B \cap \Omega$ on the partition $\Omega = A \cup A^C$.

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}(B \cap \Omega) = \mathbb{P}(B \cap (A \cup A^C)) = \mathbb{P}((B \cap A) \cup (B \cap A^C)) \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A^C \cap B) \Rightarrow \mathbb{P}(A^C \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B)\end{aligned}$$

Therefore, applying the definition of conditional probability will result in

$$\mathbb{P}(A^C|B) = \frac{\mathbb{P}(A^C \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B) - \mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} - \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = 1 - \mathbb{P}(A|B)$$

2.1.1 The (conditional) odds of an event

Generally speaking, the odds of an event can be seen as the likelihood that the event will occur relative to it not occurring.

For this reason, the odds of an event $A \subseteq \Omega$ with probability $\mathbb{P}(A) \in [0, 1)$ can be computed as

$$\mathcal{O}(A) = \frac{\mathbb{P}(A)}{\mathbb{P}(A^C)} = \frac{\mathbb{P}(A)}{1 - \mathbb{P}(A)}$$

More specifically, the conditional odds of event A given that a second event B occurred can be computed as

$$\mathcal{O}(A|B) = \frac{\mathbb{P}(A|B)}{\mathbb{P}(A^C|B)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A^C \cap B)}$$

2.1.2 The multiplication rule

The multiplication rule allows to reformulate the joint probability of two events A and B according to the following equation:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A)$$

In particular, repeatedly applying the multiplication rule to a sequence of events $\{A_1, \dots, A_n\}$ results in the chain rule, which can be formulated as

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbb{P}(A_1) \cdot \mathbb{P}(A_2|A_1) \cdot \dots \cdot \mathbb{P}(A_n|A_1 \cap \dots \cap A_{n-1})$$

2.2 Defining event independence

Two events A and B on a sample space Ω are said to be independent if knowing that B occurred does not add information about the probability of A occurring, resulting in

$$A \perp B \Leftrightarrow \mathbb{P}(A|B) = \mathbb{P}(A) \text{ and } \mathbb{P}(B|A) = \mathbb{P}(B), \text{ resulting in } \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Depending on the context, event independence can be assumed or verified, although it is always possible to show that, if A and B are events such that $\mathbb{P}(A), \mathbb{P}(B) > 0$ and $A \cap B = \emptyset$, then they events cannot be independent.

Proof:

It is possible to notice that having $A \cap B = \emptyset$ and knowing that one of the two events occurred would violate the definition of event independence because

$$\mathbb{P}(A)\mathbb{P}(B) > 0, \text{ but } \mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0.$$

2.3 Generalizing event independence

For $n > 2$ events, independence can be defined at several distinct levels.

2.3.1 Pairwise and mutual independence

A (countable) sequence of events $\{A_1, \dots, A_n\}$ is said to be pairwise independent if

$$A_i \perp A_j \quad \forall i \neq j \Leftrightarrow \mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \quad \forall i \neq j$$

Additionally, a (countable) sequence of events $\{A_1, \dots, A_n\}$ is said to be mutually independent if, for any group of k events in the sequence, it is the case that

$$\mathbb{P}\left(\bigcap_{i=1}^k A_{n_i}\right) = \prod_{i=1}^k \mathbb{P}(A_{n_i}), \text{ for } 1 \leq k \leq n \text{ and } k \in \mathbb{N}.$$

It should be noticed that, while mutual independence always implies pairwise independence, the converse statement is not true because a sequence of events can be pairwise independent but not mutually independent.

Example: A special die

Given a balanced four-faced die, which has a red face, a green face, a blue face and a striped face with all three colours, define the events

$$R = \{\text{I see red}\}, G = \{\text{I see green}\} \text{ and } B = \{\text{I see blue}\}$$

It is possible to show that these events are pairwise independent as

$$\mathbb{P}(R \cap G) = 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(G)$$

$$\mathbb{P}(R \cap B) = 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(B)$$

$$\mathbb{P}(G \cap B) = 0.25 = 0.5 \cdot 0.5 = \mathbb{P}(G)\mathbb{P}(B)$$

However, the events are not mutually independent because

$$\mathbb{P}(R \cap G \cap B) = 0.25 \neq 0.125 = 0.5 \cdot 0.5 \cdot 0.5 = \mathbb{P}(R)\mathbb{P}(G)\mathbb{P}(B)$$

2.3.2 Conditional independence

Two events A and B are said to be conditionally independent given an event C if, knowing that C occurred, observing event B does not add information about the probability of A occurring, resulting in

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C), \text{ with } \mathbb{P}(C) > 0.$$

Example: University admissions

For a student applying to a Master's Degree course, define the events

$$DS = \{\text{Accepted to Data Science}\} \text{ and } ST = \{\text{Accepted to Statistics}\}$$

Knowing that event DS occurred can provide additional information about the likelihood of event ST occurring, meaning that the two events are not independent.

However, if the admission procedure is solely based on the student's GPA C , knowing that one event occurred does not provide additional information about the other event occurring as well, making DS and ST conditionally independent given event C .

2.4 The law of total probability

Given a sample space Ω , the law of total probability allows to explain the probability of an event $A \subseteq \Omega$ by decomposing it over the partition $\Omega = B \cup B^C$, resulting in

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C)$$

Proof:

Let $A = A \cap \Omega$ on the partition $\Omega = B \cup B^C$.

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(A \cap \Omega) = \mathbb{P}(A \cap (B \cup B^C)) = \mathbb{P}((A \cap B) \cup (A \cap B^C)) \\ &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) \stackrel{\text{Multiplication rule}}{=} \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B) \end{aligned}$$

Additionally, it is possible to generalize the law of total probability to explain the probability of an event E with respect to a sequence of mutually exclusive hypotheses $\{H_1, \dots, H_n\}$ as

$$\mathbb{P}(E) = \sum_{i=1}^n \mathbb{P}(E|H_i) = \sum_{i=1}^n \mathbb{P}(E|H_i)\mathbb{P}(H_i)$$

2.5 Bayes' theorem

Bayes' theorem allows to switch between conditional probabilities by setting

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} \text{ with } \mathbb{P}(B) > 0.$$

In particular, when considering a sequence of mutually exclusive hypotheses $\{H_1, \dots, H_n\}$ with known prior probabilities, Bayes' theorem allows to compute the posterior probability of hypothesis H_i given the observed evidence E as

$$\mathbb{P}(H_i|E) = \frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\sum_{j=1}^n \mathbb{P}(E|H_j)\mathbb{P}(H_j)}$$

In fact, the posterior probability of each hypothesis given the observed evidence will be computed in terms of the likelihood $\mathbb{P}(E|H_i)$, which describes the probability of seeing the evidence given that the hypothesis H_i is true, the prior probability $\mathbb{P}(H_i)$, which describes the probability of H_i being true, and the marginal probability $\mathbb{P}(E)$, which denotes the total probability of seeing the evidence under all possible hypotheses.

Example: Iterated Bayes for medical diagnosis

Consider the events

$$T = \{\text{I test positive to a disease}\} \text{ and } D = \{\text{I actually have the disease}\}$$

In particular, the disease affects 0.1% of the population and it is known that a dedicated test correctly detects 99% of disease case but produces a false positive result on 1% of occasions, which means that

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} = \frac{0.99 \cdot 0.001}{(0.99 \cdot 0.001) + (0.01 \cdot 0.999)} = 9\%$$

Since the posterior probability appears to be quite low, Bayes' rule is applied once again, using the posterior probability as the new prior probability, resulting in

$$\mathbb{P}(D|P) = \frac{\mathbb{P}(P|D)\mathbb{P}(D)}{\mathbb{P}(P|D)\mathbb{P}(D) + \mathbb{P}(P|D^C)\mathbb{P}(D^C)} = \frac{0.99 \cdot 0.09}{(0.99 \cdot 0.09) + (0.01 \cdot 0.91)} = 91\%$$

2.5.1 The posterior odds of a hypothesis

When considering an event E on a sequence of mutually exclusive hypotheses $\{H_1, \dots, H_n\}$, it is possible to compute the posterior odds of hypothesis H_i given evidence E as

$$\mathcal{O}(H_i|E) = \frac{\mathbb{P}(H_i|E)}{\mathbb{P}(H_i^C|E)} = \frac{\frac{\mathbb{P}(E|H_i)\mathbb{P}(H_i)}{\mathbb{P}(E)}}{\frac{\mathbb{P}(E|H_i^C)\mathbb{P}(H_i^C)}{\mathbb{P}(E)}} = \frac{\mathbb{P}(E|H_i)\mathcal{O}(H_i)}{\mathbb{P}(E|H_i^C)}$$